Lecture 42

# Gauss and Stokes

**Uses.**

Summary 8: Scalar product, chain and power differentiation rules hold.
Summary 13: Cross products determine oriented perpendicular area vectors.
Summary 26: The fundamental theorem converts the integral of a partial derivative along a coordinate segment into an endpoint difference.
Summary 37: Mixed partial derivatives of a $C^2$ function agree.
Summary 38: Standing fact M supplies proper compact Fubini and admissible changes of variables.
Summary 39: Compact graph solids and spherical coordinates satisfy M under their stated boundary conditions.
Summary 40: Green holds with positive orientation on its stated graph regions; mathematical potentials are gradients.
Summary 41: Oriented surface flux uses $\boldsymbol{r}_u \times \boldsymbol{r}_v$. Divergence and curl have their Cartesian definitions; a sphere has area $4\pi R^2$.

**Objectives.** *Boundary laws compare an integral of a derivative over a region with an integral of the original field on its boundary. We prove the divergence theorem for a box and give its compact-solid extension as a sketch. We state the general Stokes theorem with its orientation convention and prove the identity on one admissible parameter patch. Explicit sphere and hemisphere calculations test the hypotheses and signs, including the effect of a singularity inside a solid. A stated conservation model then connects local density changes to boundary flux.*

By the end of this lecture you will be able to:

- Prove the box divergence theorem and relate the stated box mass-balance model to its local continuity equation.
- Check the domain of an inverse-square field and compute both boundary fluxes of a spherical shell.
- State the cited Stokes theorem with compatible surface and boundary orientations and its neighbourhood regularity hypothesis.
- Derive the single-patch Stokes identity from Green and independently check the disc and hemisphere example.

## 42.1 The divergence theorem for a box

Let $K = [a, b] \times [c, d] \times [e, f]$ with positive side lengths. Its boundary has six faces. On each face, choose the normal pointing out of the box; edges and vertices have no extra surface

contribution. For a vector field $\boldsymbol{F} = (F_1, F_2, F_3)$ that is $C^1$ on an open neighbourhood of the whole box, the formula is

$$\iint_{\partial K} \boldsymbol{F} \cdot \boldsymbol{n} \, \mathrm{d}S = \iiint_K \operatorname{div} \boldsymbol{F} \, \mathrm{d}V. \tag{42.1}$$

Here the left side is the sum over all outward-oriented faces. The notation $\partial K$ means the entire boundary, rather than only the face visible in a picture.

*Proof.* On $x = b$ the normal is $(1, 0, 0)$ and the flux is the integral of $F_1(b, y, z)$. On $x = a$ the normal is $(-1, 0, 0)$ and the flux is the negative integral of $F_1(a, y, z)$. Their sum, using the fundamental theorem and M, is

$$\int_c^d \int_e^f \big(F_1(b, y, z) - F_1(a, y, z)\big) \, \mathrm{d}z \, \mathrm{d}y$$
$$= \int_c^d \int_e^f \int_a^b \partial_x F_1(x, y, z) \, \mathrm{d}x \, \mathrm{d}z \, \mathrm{d}y = \iiint_K \partial_x F_1 \, \mathrm{d}V.$$

All derivatives are continuous on the compact box, so these proper integrals satisfy Fubini. The two $y$ faces analogously contribute $\iiint_K \partial_y F_2 \, \mathrm{d}V$, and the two $z$ faces contribute $\iiint_K \partial_z F_3 \, \mathrm{d}V$. Adding the three pairs gives Eq. (42.1) by finite linearity [P from M]. ∎

**Example 42.1** (Six faces checked separately). For $\boldsymbol{F} = (x^2, xy, z^2)$ on $K = [0, 1] \times [0, 2] \times [0, 3]$, $\operatorname{div} \boldsymbol{F} = 3x + 2z$. Thus the volume integral is

$$\int_0^1 \int_0^2 \int_0^3 (3x + 2z) \, \mathrm{d}z \, \mathrm{d}y \, \mathrm{d}x = \int_0^1 (18x + 18) \, \mathrm{d}x = 27.$$

Directly, the two $x$ faces give 6 and 0; the two $y$ faces give 3 and 0; the two $z$ faces give 18 and 0. For example, the $y = 2$ face contributes $\int_0^1 \int_0^3 2x \, \mathrm{d}z \, \mathrm{d}x = 3$. Their sum is $6 + 3 + 18 = 27$, agreeing with the proved theorem. The nonzero faces all happen to have positive coordinate normals in this example; that is a property of this field and box, not the general orientation rule.

## 42.2 The compact-solid extension and internal boundaries

The *divergence theorem*, also called *Gauss's theorem*, extends Eq. (42.1) to the following class: $K$ is a compact solid with piecewise $C^1$ boundary that can be divided into finitely many admissible compact regions between graphs, and $\boldsymbol{F}$ is $C^1$ on an open neighbourhood of all of $K$. Every component of $\partial K$ carries the normal pointing out of the solid. Balls and spherical shells belong to this class. We give the general extension as a sketch [S], not as an additional full geometric proof.

For the sketch, consider first the $F_3$ term on a graph solid $g_1(x, y) \le z \le g_2(x, y)$. The upward vector area element on its top has third component 1; the downward one on the bottom has third component $-1$. The vertical sides have horizontal normals, so their $F_3$ flux vanishes. Hence the relevant contribution is

$$\iint_D \big(F_3(x, y, g_2) - F_3(x, y, g_1)\big) \, \mathrm{d}A = \iiint_K \partial_z F_3 \, \mathrm{d}V$$

by the fundamental theorem on each vertical segment and M. The other coordinate components use their corresponding graph divisions. For a finite decomposition, each internal face is encountered twice with opposite normals, so its two fluxes cancel. Only the external

boundary components remain. The geometric choices of divisions and the treatment of their meeting edges are the part retained at sketch level. Consequences obtained through this general theorem inherit [S].

A cavity is part of the complement, so the outward normal on its boundary points into the cavity. On a spherical shell $a \le r \le b$, $0 < a < b$, the outer normal is $\boldsymbol{e}_r$ and the inner normal is $-\boldsymbol{e}_r$. Figure 42.1 shows these two boundary directions.

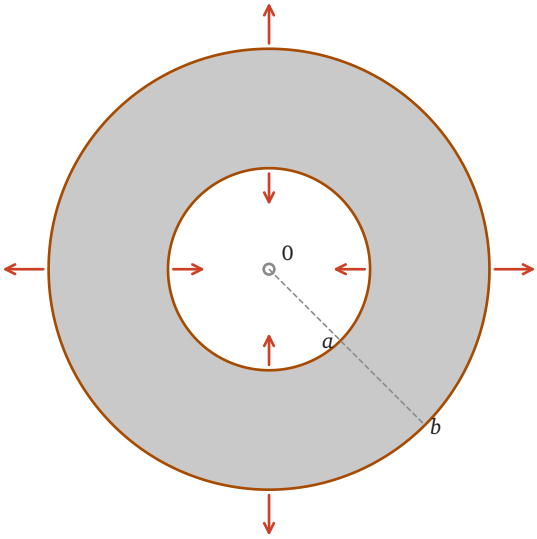



Figure 42.1: A meridian section of the shell $a \le r \le b$. Arrows show normals outward from the shell: away from the origin on the outer sphere, towards it on the inner sphere.

Choosing both normals away from the origin would orient the inner sphere incorrectly as a boundary of the shell.

## 42.3 The inverse-square field and its singularity

For $r = \|\boldsymbol{x}\| > 0$ and $\kappa \in \mathbb{R}$, consider

$$\boldsymbol{H}(\boldsymbol{x}) = \kappa \frac{\boldsymbol{x}}{r^3}.$$

Its magnitude is $|\kappa|/r^2$ and its domain excludes the origin. For $\kappa \neq 0$ it has no continuous extension there. For $\kappa = 0$ the zero field can instead be extended smoothly through the origin; no singularity claim is made about that extension. For each coordinate away from zero,

$$\partial_{x_i} H_i = \kappa(r^{-3} - 3x_i^2 r^{-5}), \qquad \operatorname{div} \boldsymbol{H} = \kappa(3r^{-3} - 3r^2 r^{-5}) = 0.$$

It is also a gradient, $\boldsymbol{H} = \nabla(-\kappa/r)$, on its punctured domain. Schwarz consequently gives zero curl there.

On a sphere of radius $R > 0$ with its usual outward normal $\boldsymbol{e}_r$, $\boldsymbol{H} \cdot \boldsymbol{e}_r = \kappa/R^2$ is constant. The sphere area formula therefore gives directly

$$\iint_{\partial B_R} \boldsymbol{H} \cdot \boldsymbol{e}_r \, \mathrm{d}S = \frac{\kappa}{R^2} 4\pi R^2 = 4\pi\kappa.$$

This is a direct surface calculation [P], independent of the general divergence theorem. For the boundary of a shell, the two contributions are

$$\text{outer flux } 4\pi\kappa, \qquad \text{inner flux } -4\pi\kappa, \qquad \text{total flux } 0.$$

The field is $C^1$ on a neighbourhood of the entire shell. The general divergence theorem [S] is applicable and agrees with zero volume integral of the divergence. For $\kappa \neq 0$, applying it to the solid ball instead would be invalid: the field is not defined, let alone $C^1$, near the whole ball. The vanishing divergence on the punctured domain does not imply zero flux through its surrounding sphere.

The same comparison explains a wider statement with its hypotheses. Let a compact solid $K$ of the stated Gauss class contain a closed ball $B_a$ centred at the origin strictly inside its interior, and assume $K \setminus \operatorname{int} B_a$ also admits that finite graph decomposition. Apply the sketch theorem to this punctured solid: its divergence integral is zero and the added inner sphere contributes $-4\pi\kappa$. Thus the total outward flux through the original boundary $\partial K$ is $4\pi\kappa$ [S]. This version requires both the solid and the punctured solid to satisfy the stated geometric conditions. It is not a claim about an arbitrary surface without those conditions.

## 42.4 Stokes's theorem and the induced boundary direction

An oriented surface has a chosen normal; its boundary direction must be chosen compatibly. Locally, traversing the boundary with the surface on the left and the chosen normal pointing upwards describes the convention. Equivalently, the right-hand rule relates the positive normal to the positive boundary direction. For an upward planar disc this direction is anticlockwise as seen from above. For several boundary components, this rule applies to each component with the same chosen surface orientation.

**Theorem 42.2** (Stokes's theorem, cited)**.** *Let $S$ be a compact oriented piecewise $C^2$ surface with piecewise $C^1$ boundary, with finitely many regular parametrised pieces and compatible normals. Let $\boldsymbol{F}$ be $C^1$ on an open neighbourhood of $S$. Orient every component of $\partial S$ by the induced boundary convention. Then*

$$\oint_{\partial S} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x} = \iint_S \operatorname{curl} \boldsymbol{F} \cdot \boldsymbol{n} \, \mathrm{d}S.$$

The general assertion is cited [C]: Stewart, §16.8. For finitely many boundary components, including an empty boundary, an accessible primary statement is J. Stanfill, *Calculus III Lecture Notes, Baylor*, §30, Theorem 30.1 and its following closed-surface note, printed p. 147. Its preceding paragraph permits finite unions of patches and boundary exceptions. Our piecewise $C^2$ and neighbourhood $C^1$ hypotheses are sufficient for that statement. The single-boundary-curve form is also B. Poonen, MIT 18.02 notes, Fall 2021, Theorem 19.1, printed p. 133. No smoothness throughout a solid enclosed by $S$ is requested by Stokes; the field must be $C^1$ near the whole surface itself. A singular point on that surface would invalidate the hypothesis.

## 42.5 A proof on one admissible parameter patch

There is a useful proved case behind the general cited formula. Let $S$ be a regular $C^2$ patch $\boldsymbol{r}(u, v)$ over a compact region $E$ where the proved Green theorem applies. Assume

its boundary map gives the surface boundary without extra identifications, and orient $S$ by $\boldsymbol{r}_u \times \boldsymbol{r}_v$ and $\partial E$ positively. The induced surface boundary is the image of this positive parameter boundary. Define the two continuously differentiable scalar functions

$$A(u,v) = \boldsymbol{F}(\boldsymbol{r}) \cdot \boldsymbol{r}_u, \qquad B(u,v) = \boldsymbol{F}(\boldsymbol{r}) \cdot \boldsymbol{r}_v.$$

The chain rule for the boundary parametrisation gives $\oint_{\partial S} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x} = \oint_{\partial E} A\,\mathrm{d}u + B\,\mathrm{d}v$. Green converts this to $\iint_E (\partial_u B - \partial_v A)\,\mathrm{d}u\,\mathrm{d}v$. When differentiating $A, B$, the terms involving second derivatives of $\boldsymbol{r}$ cancel by Schwarz, since $\boldsymbol{r}$ is $C^2$. Writing $\boldsymbol{r} = (x, y, z)$, the remaining expression is

$$\begin{aligned} \partial_u B - \partial_v A &= (\partial_x F_2 - \partial_y F_1)(x_u y_v - x_v y_u) \\ &\quad + (\partial_y F_3 - \partial_z F_2)(y_u z_v - y_v z_u) \\ &\quad + (\partial_z F_1 - \partial_x F_3)(z_u x_v - z_v x_u) \\ &= \operatorname{curl} \boldsymbol{F}(\boldsymbol{r}) \cdot (\boldsymbol{r}_u \times \boldsymbol{r}_v). \end{aligned}$$

The derivatives of the field in this display are evaluated at $\boldsymbol{r}(u,v)$. The equality follows by grouping each pair of distinct coordinate indices; the equal-index terms cancel directly. The resulting integral is exactly the surface flux definition in Lecture 41, proving Stokes in this single-patch setting [P from M and Green]. This argument does not by itself prove the general geometric patching assertion in Theorem 42.2, which retains [C].

## 42.6 The same boundary on a disc and a hemisphere

Let $\boldsymbol{F} = (-y/2, x/2, 0)$, which is $C^1$ everywhere and has $\operatorname{curl} \boldsymbol{F} = (0, 0, 1)$. Take the unit disc in $z = 0$ with upward normal and the upper unit hemisphere with outward normal. Both induce the same boundary circle direction: anticlockwise when viewed from positive $z$. For the disc this follows from the planar convention. For the hemisphere, at the point $(1, 0, 0)$ the outward normal points in positive $x$, while the tangent into the hemisphere points in positive $z$; the compatible boundary tangent points in positive $y$, giving the same turn.

Directly parametrise the circle by $(\cos t, \sin t, 0)$, $0 \le t \le 2\pi$. The field's scalar product with its velocity is $1/2$, so

$$\oint_{\partial S} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x} = \pi.$$

On the upward disc, the curl's normal component is 1, and its area is $\pi$, giving flux $\pi$ as well. On the hemisphere use $0 \le \theta \le \pi/2$ and one azimuthal turn. The outward vector area element is $\sin\theta\, \boldsymbol{e}_r$ by Lecture 41. Thus

$$\iint_{S_{\mathrm{hem}}} \operatorname{curl} \boldsymbol{F} \cdot \boldsymbol{n}\,\mathrm{d}S = \int_0^{2\pi} \int_0^{\pi/2} \cos\theta \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi = \pi.$$

These three explicit calculations are proved [P] independently of the cited general theorem, and verify its agreement for both surfaces. No claim that the integrands or normal vectors agree point by point is needed: the two surface integrals have the same oriented boundary value. Reversing both surface and boundary orientations reverses every value.

## 42.7 Consequences and a conservation model

The boundary laws also explain when one integration surface may be replaced by another. Suppose two surfaces satisfy Stokes's hypotheses for the same field, have the same boundary

curves, and induce the same direction on every boundary component. Applying the cited theorem to each gives

$$\iint_{S_1} \operatorname{curl} \boldsymbol{F} \cdot \boldsymbol{n}_1 \, \mathrm{d}S = \oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x} = \iint_{S_2} \operatorname{curl} \boldsymbol{F} \cdot \boldsymbol{n}_2 \, \mathrm{d}S.$$

This surface-replacement conclusion inherits [C]. The field must be $C^1$ near both surfaces; it is not enough to check only the common boundary. No region between the surfaces is needed for this argument. For a closed surface the boundary is empty, so Stokes gives zero total flux of a curl [C]. This applies only when that curl comes from a field with the specified neighbourhood regularity. A field that merely has zero divergence on a punctured domain need not be such a curl near a whole surrounding sphere, as Sheet 42 asks you to show.

There is a corresponding scalar consequence of Gauss. If $v$ is $C^2$ near a box or a solid of the stated Gauss class, then $\nabla v$ is $C^1$ there. Its normal component is called the *outward normal derivative*, $\partial_n v = \nabla v \cdot \boldsymbol{n}$. Gauss gives

$$\iint_{\partial K} \partial_n v \, \mathrm{d}S = \iiint_K \Delta v \, \mathrm{d}V.$$

The formula is [P] for a box and [S] for the general solid class, exactly like Gauss. In particular, a harmonic function has zero total outward normal derivative under those hypotheses. This is a statement about the sum of the boundary contributions, not a pointwise claim that its normal derivative vanishes everywhere.

For a physical application, let $\rho(t, x) \geq 0$ be mass per unit volume and let $\boldsymbol{J}(t, x)$ be mass flux per unit area per unit time. Assume both are $C^1$ jointly on an open time–space domain. Fix a time and a spatial box whose neighbourhood stays in that domain for a small time interval. The model of mass conservation on this fixed box is the assumption

$$\frac{\mathrm{d}}{\mathrm{d}t} \iiint_K \rho(t, x) \, \mathrm{d}V = - \iint_{\partial K} \boldsymbol{J}(t, x) \cdot \boldsymbol{n} \, \mathrm{d}S. \tag{42.2}$$

Positive outward flux removes mass from the box, which fixes the minus sign. This is a physical premise, not a mathematical theorem about every pair of functions $\rho, \boldsymbol{J}$.

The parameter clause of M applies on compact time subintervals and this fixed compact box because of joint $C^1$ regularity on a neighbourhood. It moves the time derivative inside the proper integral. The proved box divergence theorem then turns the assumed balance into

$$\iiint_K \left( \partial_t \rho + \operatorname{div}_x \boldsymbol{J} \right) \mathrm{d}V = 0.$$

The subscript $x$ means divergence in the three spatial coordinates. If the model balance holds for every sufficiently small spatial box, this integral identity implies the pointwise *continuity equation*

$$\partial_t \rho + \operatorname{div}_x \boldsymbol{J} = 0. \tag{42.3}$$

Here is the full localisation argument. At a fixed time the integrand is continuous. If its value at one point were positive, continuity would make it at least half that positive value on a sufficiently small box around the point. Its integral would be positive, contradicting the identity. A negative value gives the same contradiction with reversed sign. Hence it vanishes at every point. Conversely, integrate the pointwise equation on any such box, apply M's parameter rule and the box divergence theorem in reverse order, and recover the model balance. Thus the equivalence is proved [P from M] under the stated smoothness and all-small-box assumption.

When a smooth velocity field $\boldsymbol{u}$ carries the mass, the additional transport model is $\boldsymbol{J} = \rho\boldsymbol{u}$. The scalar product rule in each coordinate rewrites the continuity equation as

$$\partial_t \rho + \boldsymbol{u} \cdot \nabla_x \rho + \rho \operatorname{div}_x \boldsymbol{u} = 0.$$

For a constant positive density, it reduces to $\operatorname{div}_x \boldsymbol{u} = 0$. Zero total flux here does not require the velocity or every individual face flux to vanish.

For example, choose $\rho_0 > 0$ and $\alpha \in \mathbb{R}$, and set on all time and space

$$\boldsymbol{u}(t, x) = \alpha x, \qquad \rho(t, x) = \rho_0 e^{-3\alpha t}, \qquad \boldsymbol{J} = \rho\boldsymbol{u}.$$

These functions are smooth and the density is always positive. The density has time derivative $-3\alpha\rho$, spatial gradient zero, and the velocity has divergence $3\alpha$. They satisfy the continuity equation exactly. In a fixed box of volume $V_K$, the mass is $\rho(t)V_K$ and its rate of change is $-3\alpha\rho(t)V_K$; the outward flux is $3\alpha\rho(t)V_K$ by the proved box theorem. For $\alpha > 0$ the flow expands and the fixed-box mass decreases; for $\alpha < 0$ it increases; for $\alpha = 0$ the velocity vanishes and the mass stays constant. All three cases agree with the same balance law. This example connects a local derivative equation to a measurable boundary flow while retaining the model assumptions that justify it.

**Summary 42.**

- Gauss equates total outward boundary flux to the volume integral of divergence for a box [P]. The stated compact-solid finite-graph extension is [S]; its field must be $C^1$ near the entire solid.
- Every boundary component is outward from the solid. On a shell this means a radially inward normal on the inner sphere. An inverse-square singularity prevents using Gauss on a ball containing it when $\kappa \neq 0$.
- General Stokes [C] requires the stated compact oriented surface, compatible boundary orientation and a field $C^1$ near the surface. Its single admissible $C^2$ patch case follows from Green [P].
- Under the stated joint $C^1$ hypotheses, mass balance on every small box is equivalent to the continuity equation Eq. (42.3) [P from M], with the physical balance law retained as an assumption.
- The disc and upper hemisphere calculations both give $\pi$ for the stated field and common induced circle direction. These are direct checks, rather than a proof of the general cited theorem.

Sheet 42 applies the boundary laws while checking normals, holes, singularities and the dependence of each route on proved, sketched or cited inputs. This completes the course's passage from limits and local linear approximations to integral relations over regions and boundaries.

## References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§16.8–16.9. B. Poonen, *18.02 Lecture Notes*, MIT, Fall 2021, Theorem 19.1, printed p. 133, author's course notes [C]. J. Stanfill, *Calculus III Lecture Notes, Baylor*, MTH 2321, §30, Theorem 30.1 and closed-surface note, printed p. 147, author's course notes [C].